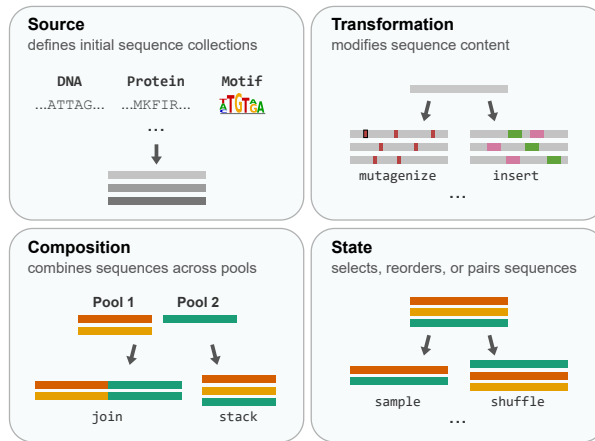
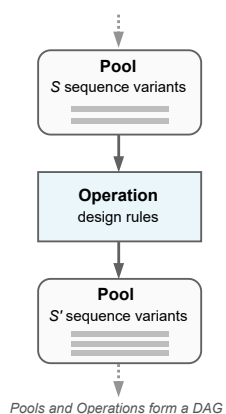
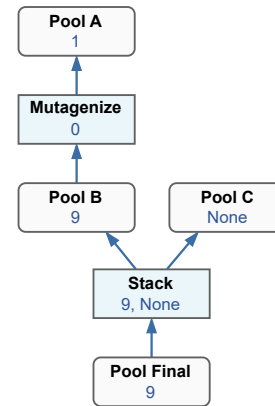


A

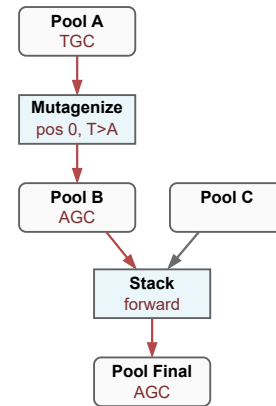


B

State assignment

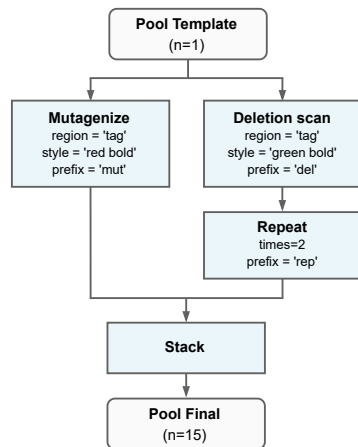


Sequence construction



C

Template: TCCGACT<tag>GCA</tag>ATTCGGA



Design DataFrame

state	name	stack		mutagenize		del_scan		repeat
		branch	pos	sub	pos	base	idx	
0	mut_0	0	0	G>A	None	None	None	
1	mut_1	0	0	G>C	None	None	None	
2	mut_2	0	0	G>T	None	None	None	
3	mut_3	0	1	C>A	None	None	None	
4	mut_4	0	1	C>G	None	None	None	
5	mut_5	0	1	C>T	None	None	None	
6	mut_6	0	2	A>C	None	None	None	
7	mut_7	0	2	A>G	None	None	None	
8	mut_8	0	2	A>T	None	None	None	
9	del_0.rep_0	1	None	None	0	G	0	
10	del_0.rep_1	1	None	None	0	G	1	
11	del_1.rep_0	1	None	None	1	C	0	
12	del_1.rep_1	1	None	None	1	C	1	
13	del_2.rep_0	1	None	None	2	A	0	
14	del_2.rep_1	1	None	None	2	A	1	

```

>>> pool_final.print_library(show_name=True)
name      seq
mut_0     TCCGACT<tag>ACA</tag>ATTCGGA
mut_1     TCCGACT<tag>CCA</tag>ATTCGGA
mut_2     TCCGACT<tag>TCA</tag>ATTCGGA
mut_3     TCCGACT<tag>GAA</tag>ATTCGGA
mut_4     TCCGACT<tag>GGA</tag>ATTCGGA
mut_5     TCCGACT<tag>GTA</tag>ATTCGGA
mut_6     TCCGACT<tag>GCC</tag>ATTCGGA
mut_7     TCCGACT<tag>GCG</tag>ATTCGGA
mut_8     TCCGACT<tag>GCT</tag>ATTCGGA
del_0.rep_0 TCCGACT<tag>-CA</tag>ATTCGGA
del_0.rep_1 TCCGACT<tag>-CA</tag>ATTCGGA
del_1.rep_0 TCCGACT<tag>G-A</tag>ATTCGGA
del_1.rep_1 TCCGACT<tag>G-A</tag>ATTCGGA
del_2.rep_0 TCCGACT<tag>GC-</tag>ATTCGGA
del_2.rep_1 TCCGACT<tag>GC-</tag>ATTCGGA
  
```

None = inactive branch ■ mutation ■ deletion ■ tagged region